Peizhi Shi

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Research Fellow, EPSRC Future Advanced Metrology Hub
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Research Interests

Machine learning, computer vision, intelligent manufacturing

Education

University of Manchester — Manchester, UK
PhD in Computer Science — 2013 – 2019
Supervisor: Dr. Ke Chen

University of Science and Technology of China — Hefei, China
Master in Software Engineering — 2010 – 2013

Guilin University of Electronic Technology — Guilin, China
Bachelor in Computer Science — 2006 – 2010

Work Experience

EPSRC Future Advanced Metrology Hub
Research Fellow — 2019 – Present

Teaching Experience

Algorithms and Imperative Programming
Teaching Assistant, University of Manchester — 2013, 2014, 2015, 2017

Modelling and Visualization of High Dimensional Data
Teaching Assistant, University of Manchester — 2015

Introduction to Machine Learning
Teaching Assistant, University of Manchester — 2015

3rd Asian-Pacific Summer School on Formal Methods
Teaching Assistant, University of Science and Technology of China — 2011

Publications

Intersecting machining feature localisation and recognition via single shot multibox detector
Peizhi Shi, Qunfen Qi, Yuchu Qin, Paul Scott, Xiangqian Jiang.
IEEE Transactions on Industrial Informatics, 2021.

Status, issues, and future of computer-aided part orientation for additive manufacturing
Yuchu Qin, Qunfen Qi, **Peizhi Shi**, Paul J Scott, Xiangqian Jiang.
The International Journal of Advanced Manufacturing Technology, 2021.

Automatic determination of part build orientation for laser powder bed fusion
Yuchu Qin, Qunfen Qi, **Peizhi Shi**, Paul J Scott, Xiangqian Jiang.

Virtual and Physical Prototyping, 2021.

Multiple-attribute decision-making based on picture fuzzy archimedean power Maclaurin symmetric mean operators
Yuchu Qin, Xiaolan Cui, Meifa Huang, Yanru Zhong, Zhemin Tang, **Peizhi Shi**.
Granular Computing, 2021.

A novel learning-based feature recognition method using multiple sectional view representation
Peizhi Shi, Qunfen Qi, Yuchu Qin, Paul Scott, Xiangqian Jiang.
Journal of Intelligent Manufacturing, 2020.

Novel operational laws and power muirhead mean operators of picture fuzzy values in the framework of Dempster-Shafer theory for multiple criteria decision making
Yuchu Qin, Qunfen Qi, **Peizhi Shi**, Paul J Scott, Xiangqian Jiang.
Computers & Industrial Engineering, 2020.

Automatic generation of alternative build orientations for laser powder bed fusion based on facet clustering
Yuchu Qin, Qunfen Qi, **Peizhi Shi**, Paul J Scott, Xiangqian Jiang.
Virtual and Physical Prototyping, 2020.

Linguistic interval-valued intuitionistic fuzzy archimedean prioritised aggregation operators for multi-criteria decision making
Yuchu Qin, Qunfen Qi, **Peizhi Shi**, Paul J Scott, Xiangqian Jiang.
Journal of Intelligent & Fuzzy Systems, 2020.

Linguistic interval-valued intuitionistic fuzzy archimedean power muirhead mean operators for multiattribute group decision-making
Yuchu Qin, Xiaolan Cui, Meifa Huang, Yanru Zhong, Zhemin Tang, **Peizhi Shi**.
Complexity, 2020.

Archimedean muirhead aggregation operators of q-rung orthopair fuzzy numbers for multicriteria group decision making
Yuchu Qin, Xiaolan Cui, Meifa Huang, Yanru Zhong, Zhemin Tang, **Peizhi Shi**.
Complexity, 2019.

Learning constructive primitives for real-time dynamic difficulty adjustment in super mario bros
Peizhi Shi, Ke Chen.
IEEE Transactions on Games, 2017.

Online level generation in super mario bros via learning constructive primitives
Peizhi Shi, Ke Chen.
IEEE Conference on Computational Intelligence and Games, 2016.

Modular verification of concurrent thread management
Yu Guo, Xinyu Feng, Zhong Shao, **Peizhi Shi**.
Asian Symposium on Programming Languages and Systems, 2012.

Skills

Machine Learning
Deep learning, active learning, cost-sensitive classification, random forests, support vector machine, neural network, hidden markov model, naive bayes, principal component analysis, liner regression, softmax, clustering, graph partitioning, manifold learning, Q learning and Thompson sampling.

Computer Science and Mathematics
Calculus, linear algebra, statistics, data structure, algorithm, compiling theory, operating system, database, programming (i.e. C, C++, Python, Matlab, Java, C#, assembly language).

Languages
Chinese (native), English (proficient)